

The Kelman logo, featuring the word "kelman" in a blue sans-serif font, followed by a blue circular icon containing a stylized white arrow pointing upwards and to the right.

kelman

A photograph of a yellow rugged case with its lid open, revealing a portable gas analyzer. A person's hand is touching the device's screen. In the background, there is a large industrial transformer at a power substation.

**TRANSPORT X**  
Portable Dissolved  
Gas Analysis

# TRANSPORT X

Accurate knowledge of the condition of transformers is essential for all electrical networks. This information allows valuable assets to be maximised and expensive failures to be avoided.

Dissolved Gas Analysis (DGA) is an established technique and is recognised as the most important test in monitoring power transformers. It is now being successfully extended to other oil filled equipment such as tap changers and circuit breakers.

**TRANSPORT X** uses new technology to give accurate, reliable results in a matter of minutes. **TRANSPORT X** represents an invaluable tool for Asset Management and will increase the power of any DGA program.



## Features & Benefits

- **Accurate & Sensitive**  
**TRANSPORT X** has a wide detection range with excellent accuracy for all 7 fault gases.
- **Moisture Analysis**  
**TRANSPORT X** measures water content in oil and can express concentration as parts per million or relative saturation.
- **Simple**  
Extremely easy step-by-step operation. No extensive calibration, set-up or interpretation of results required.
- **Fast**  
On-site results in a matter of minutes.
- **No Consumables**  
**TRANSPORT X** requires no calibration gases or carrier gases.
- **Portable & Rugged**  
11kg (24lbs) in a rugged convenient carry case.
- **DGA Diagnostics**  
Includes DGA diagnostic algorithms - Rogers' Ratios, Duval's Triangle and IEEE Key gas. Also user settable *Caution* and *Warning* thresholds on all gases.
- **Test Gas Samples**  
Ability to test gas samples taken from Buchholz Relays.
- **PC Software**  
Includes **TransportPro** PC software package to allow storage, exporting and subsequent trending and analysis of results.
- **Extensive Worldwide User List.**
- **PERCEPTION**  
Complimentary Kelman **PERCEPTION** software with **TRANSPORT X** for trending & analysis of results.

# PORTABLE DISSOLVED GAS ANALYSIS



**TRANSPORT X** uses state of the art technology to achieve unprecedented performance for DGA equipment. Extensive field and laboratory use worldwide has proven the **TRANSPORT X** gives highly reliable results and the unit is genuinely suitable for field conditions.

**TRANSPORT X** minimises the risk of carry-over between tests. With the ability to go from high gassed samples (such as tap changers) to low gassed samples (such as main tanks) with no contamination of results the user can confidently test all types of oil filled equipment.

Internal diagnostic software helps to translate ppm data into useful information. Established algorithms assist the user to analyse the condition of the transformer. The accompanying **TransportPro** PC software allows the user to download records for trending and further diagnosis.

**TRANSPORT X** represents a new generation of DGA test equipment: accurate, portable and simple to use.

### Additional Features

- Records compatible with **PERCEPTION** from Kelman and TOA (from DeltaX Research).
- Up to 20,000 records stored in internal memory.
- Embedded thermal printer gives hard copy of results on-site.
- Touch screen controls for easy operation.

### Additional Options

- 'System Check Kit' for verification of unit operation.
- Gas test kit allows analysis of buchholz gas samples.
- Transit Case. Provides extra protection for air travel and transportation
- Sample Cooler. Allows hot oil samples to be cooled for testing in minutes.

### Related Products

- TRANSFIX** - 8 gas & moisture monitor for transformer main tank.
- TAPTRANS** - Complete transformer DGA monitoring for both main tank & LTC.
- MINITRANS** - 3 gas & moisture monitor for transformer main tank.

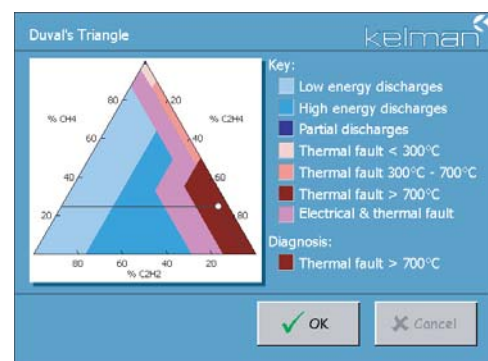
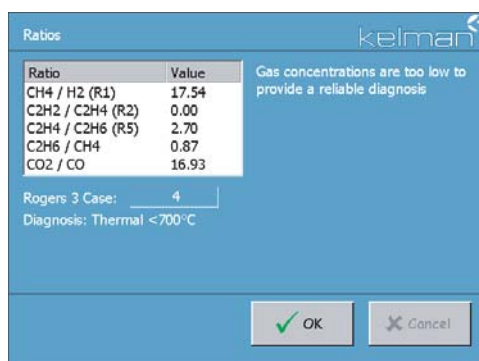
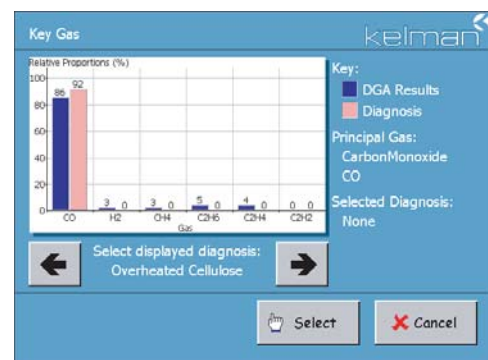
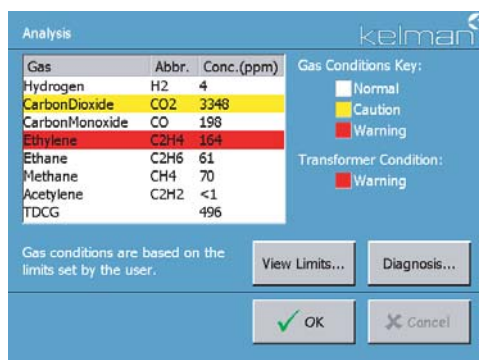


Fig 1 - Sample displays from TRANSPORT X DGA diagnostics.

# TECHNICAL SPECIFICATION

| PARAMETER                                  | VALUE/MEETS                                                   |
|--------------------------------------------|---------------------------------------------------------------|
| COMPOUND                                   | MEASUREMENT RANGE (ppm)                                       |
| HYDROGEN (H <sub>2</sub> )                 | 5 - 5,000                                                     |
| CARBON MONOXIDE (CO)                       | 1 - 50,000                                                    |
| CARBON DIOXIDE (CO <sub>2</sub> )          | 2 - 50,000                                                    |
| METHANE (CH <sub>4</sub> )                 | 1 - 50,000                                                    |
| ACETYLENE (C <sub>2</sub> H <sub>2</sub> ) | 1 - 50,000                                                    |
| ETHANE (C <sub>2</sub> H <sub>6</sub> )    | 1 - 50,000                                                    |
| ETHYLENE (C <sub>2</sub> H <sub>4</sub> )  | 1 - 50,000                                                    |
| WATER (H <sub>2</sub> O)                   | ± 3 ppm                                                       |
|                                            | (Note: Buchholz gas samples<br>LDL is 100 ppm for all gases). |
| ACCURACY                                   | ± 5% or ± 2 ppm<br>(whichever is greater)                     |
| TEMPERATURE RANGE                          | 0 - 50°C operating<br>10 - 30°C ambient                       |
| POWER SUPPLY                               | 100 - 250 Vac; 47 - 63Hz; 12 W                                |
| OUTPUT DIGITAL                             | USB                                                           |
| OUTPUT HARDCOPY                            | 2" THERMAL PRINTER                                            |
| WEIGHT                                     | 11 KG (24 lbs) (unit only)                                    |
| SIZE                                       | 170 X 340 X 460 mm                                            |
| OIL SAMPLE VOLUME                          | 50 ml                                                         |
| GAS SAMPLE VOLUME                          | 5 ml                                                          |



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